

Numerical Computing II
Assignment III

Can we make Numerical Differentiation more accurate?

Investigation of Richardson Extrapolation

In the previous assignment, you compared Forward, Backward and Central Difference methods. You found that Central Difference has an error approximately proportional to h^2 . In this assignment, you will investigate whether Richardson Extrapolation can remove the leading h^2 error and produce a higher-order approximation.

Implement the following numerical differentiation methods:

- 1 Central Difference
- 2 Richardson Extrapolation

Compare their performance using following test functions.

- $f(x) = e^x$
- $f(x) = \sin x$
- $f(x) = \cos x$
- $f(x) = x^3 - 2x + 1$
- Use at least six different values of h (e.g., 10^{-1} , 10^{-2} , ..., 10^{-6})
- Extend the calculation to $h=10^{-7}$, 10^{-8} , ... and investigate when round-off error begins to dominate.
- For each h , calculate:
 - 1 Central Difference $D(h)$
 - 2 Richardson approximation $R(h)$
 - 3 Exact derivative
 - 4 Absolute error of $D(h)$
 - 5 Absolute error of $R(h)$
- Evaluation Point: Approximate $f'(1)$ for all three functions.
- Compute: $Absolute\ Error = |f'_{exact} - f'_{numerical}|$

Prepare and submit:

- Plots of the functions listed
- Algorithm/pseudocode
- A results table
- Convergence Investigation: a log-log plot, determine the slope of each curve and compare your observed slopes with the theoretical orders: $D(h) = O(h^2)$, $R(h) = O(h^4)$.
- Answer the following:
 - Q1. Does Richardson always give a smaller error than Central Difference?
 - Q2. By approximately what factor does the error decrease when h is reduced by 10?
 - Q3. Does Richardson continue to improve as h becomes very small?
 - Q4. Does the log-log slope support the theoretical prediction?

- Q5. Does Richardson achieve the expected $O(h^4)$ behavior?
- Q6. What role does floating-point round-off error play?